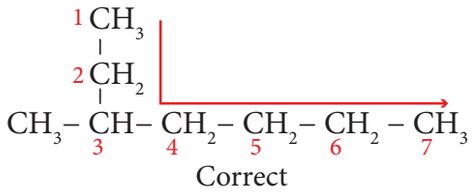
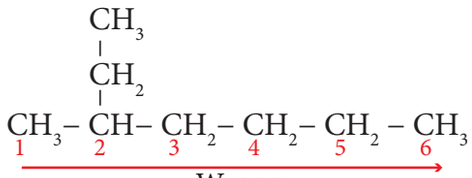
carbon chain is numbered from the end, which is closest to substituent. (Rule 2)



Correct



Wrong

Step 3: All are single bonds between the carbon atoms and thus the suffix is 'ane'.

Step 4: The substituent is a methyl group and it is located at third carbon. So, its locant number is 3. Thus the prefix is '3-Methyl'. (Rule 6)

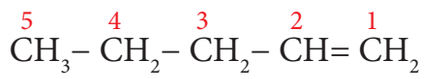
Hence the name of the compound is **3-Methyl + hept + ane = 3 –Methylheptane**

Example 4: CH_3-CH_2-CH_2-CH=CH_2

Step 1: It is a 'five- carbon atoms chain' and hence the root word is 'Pent'. (Rule 1)

Step 2: There is a carbon to carbon double bond. The suffix is 'ene'.

Step 3: The carbon chain is numbered from the end such that double bond has the lowest locant number as shown below: (Rule 3):



Step 4: The locant number of the double bond is 1 and thus the suffix is '-1-ene'.

So, the name of the compound is **Pent + (-1-ene) = Pent-1-ene**

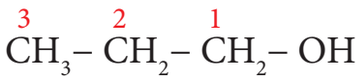
11.5.5 IUPAC Nomenclature of other classes – Solved examples

Example 1: CH_3-CH_2-CH_2-OH

Step1: The parent chain consists of 3 carbon atoms. The root word is 'Prop'.

Step 2: There are single bonds between the carbon atoms of the chain. So, the **primary suffix** is 'ane'.

Step 3: Since, the compound contains –OH group, it is an alcohol. The carbon chain is numbered from the end which is closest to –OH group. (Rule 3)



Step 4: The locant number of –OH group is 1 and thus the secondary suffix is '1-ol'.

The name of the compound is **Prop + ane + (1-ol) = Propan-1-ol**

Note: Terminal 'e' of 'ane' is removed as per Rule 5

Example 2: CH_3COOH

Step1: The parent chain consists of 2 carbon atoms. The root word is 'Eth'.

Step 2: All are single bonds between the carbon atoms of the chain. So the primary suffix is 'ane'.

Step 3: Since the compound contains the–COOH group, it is a carboxylic acid. The secondary suffix is 'oic acid'

The name of the compound is **Eth + ane + oic acid) = Ethanoic acid**

Table 11.6 lists IUPAC names homologs of various classes of organic compounds

Test yourself:

Obtain the IUPAC name of the following compounds systematically:

(a) CH_3CHO

(b) $CH_3CH_2COCH_3$

(c) $ClCH_2$-CH_2-CH_2-CH_3

10th_Science_Unit-11.indd 162 06-12-2021 18:38:53